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Measuring Attentiveness in Self-Administered Surveys

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Abstract The surge in online self-administered surveys has given rise to an extensive body of literature on respondent inattention, also known as careless or insufficient effort responding. This burgeoning literature has outlined the consequences of inattention and made important strides in developing effective methods to identify inattentive respondents. However, differences in terminology, as well as a multiplicity of different methods for measuring and correcting for inattention, have made this literature unwieldy. We present an overview of the current state of this literature, highlighting commonalities, emphasizing key debates, and outlining open questions deserving of future research. Additionally, we emphasize the key considerations that survey researchers should take into account when measuring attention.

Introduction

Over the past twenty years, online surveys have exploded in popularity, quickly becoming the preferred survey mode for public opinion researchers, academics, and practitioners alike. These online surveys are typically conducted in an unsupervised manner, allowing respondents to more easily complete the survey. But perhaps taking a survey has become too easy. The

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surge in online self-administered surveys has given rise to an extensive body of literature on survey inattention, also known as careless or insufficient effort responding. Inattention is widely recognized as a significant problem for public opinion research. Inattentive respondents can introduce noise to the data, generating additional measurement errors that can affect scale reliability (Huang et al. 2015; Silber, Danner, and Rammstedt 2019; Pyo and Maxfield 2021) and estimated treatment effects (Berinsky, Margolis, and Sances 2014; Maniaci and Rogge 2014; Hauser and Schwarz 2015; Abbey and Meloy 2017; DeSimone and Harms 2018; Brühlmann et al. 2020).

This burgeoning literature outlines the consequences of inattention and has made important strides in developing effective methods to identify inattentive respondents. But it has also become unwieldy. As a result, it is often difficult for public opinion researchers to get a handle on current research, much less implement best practices. This paper presents an overview of the current state of the literature, emphasizing central debates and open questions deserving of future research. Additionally, it emphasizes the key considerations that survey researchers should take into account when measuring attention. We acknowledge the dynamic nature of survey research and aim to offer researchers a flexible framework for considering survey attention, independent of specific measurement strategies. While the bulk of the studies included in this review rely on online surveys, the lessons about survey attention are widely applicable to various self-administered survey formats.

We begin by providing some background on what constitutes survey inattention, triangulating differences in terminology used by scholars across fields who draw from the same theoretical frameworks. We then provide an overview of the two general approaches to measuring attention—direct and indirect measures[1]—and examine the challenges associated with these strategies. Next, we discuss the importance of using well-validated measurement strategies and how a multitude of factors, including both researcher decisions and respondent characteristics, can influence attention levels. Finally, we highlight lingering questions regarding how to handle inattentive respondents and discuss potential risks associated with measuring attention.

There are two general themes we hope to convey in this review. The first is the importance of transparency when using attention measurement strategies. Researchers first make decisions about how to measure attentiveness and then decide what to do with inattentive respondents. It is crucial for researchers to disclose any and all research decisions that affect the size and makeup of the sample. Second, researchers must adopt a holistic approach when measuring attention, factoring in the underlying causes of inattention within their survey and recognizing the potential ramifications of their

1. Throughout the paper we discuss various direct and indirect measures. For detailed explanations, applications, and citations of specific attentiveness measures, please refer to tables 2 and 3.

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chosen measurement approach on the final results. Importantly, addressing inattentive responding does not have a singular "solution," and this paper will not put one forward. Rather, our focus is on acknowledging inattentive responding as an inherent feature of self-administered surveys that researchers can minimize through considerate survey design, identify using different approaches, and manage once the data is collected. Our goal is therefore to provide researchers with a nuanced understanding of the inattentiveness literature and the ability to apply methodologies that align with their specific research goals.

Theory of Survey Inattention

Much of the research on survey inattention relies on existing theories of survey response behavior to conceptualize inattentive responding and understand the inferential threats posed by this type of behavior. While not all articles do so explicitly, the theoretical frameworks used by scholars to understand inattentive responding have roots in the theories of satisficing (Krosnick 1991) and the cognitive model of survey response (Tourangeau, Rips, and Rasinski 2000). Inattention can be conceptualized as a subset of extreme satisficing behaviors where respondents may not even successfully complete the comprehension step of the response process when responding to a survey (Anduiza and Galais 2017). As we discuss in greater length in a later section, this conceptualization of inattention accounts for both individual ability and motivation as well as external features, including survey design and survey-taking context. Additionally, it is important to clarify that inattention is not a dichotomous concept where one is either attentive or inattentive. In a later section, we highlight how the use of multiple indicators to measure respondent inattention can be leveraged to create a more granular index of response inattention.

In table 1, we present a list of the various terms and definitions used across fields. Although scholars from different fields draw from the same theoretical frameworks when investigating survey inattention, there are terminological differences in how this type of behavior is discussed. Terms such as "inattentive responding," "insufficient effort responding," and "careless responding" are all used by researchers working in this space. For instance, Oppenheimer, Meyvis, and Davidenko's (2009) seminal piece introducing instructional manipulation checks (IMCs) explicitly frames that strategy as a method to detect satisficing. Political scientists Berinsky, Margolis, and Sances (2014) build on Oppenheimer, Meyvis, and Davidenko's (2009) piece and categorize this same behavior as "inattentive responding," which they define as "respondents who offer careless or haphazard survey responses" (p. 741). Meade and Craig (2012) use "careless responding" to refer to a subset of response bias termed content

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Table 1. Definitions of attentiveness used throughout disciplines.

Term definition	Citations
Inattentive/low-effort responding Low-quality responses from participants who devote less effort and offer quick answers.	• Bowling et al. (2016)[P] • Abbey and Meloy (2017)[G] • Buchanan and Scofield (2018)[P] • Malone and Lusk (2018)[B] • Breitsohl and Steidelmüller (2018)[P] • Olamijuwon (2021)[S] • Alvarez and Li (2023)[G] • Kane, Velez, and Barabas (2023)[G]
Insufficient effort responding (IER) "IER may manifest itself as either random responding or as non-random response patterns, and it encompasses both unintentional, occasional careless responding, and intentional 'speeding-through' of survey items" (Huang et al. 2015, p. 301).	• Huang et al. (2012; 2015)[P] • Iaconelli and Wolters (2020)[P] • Bowling et al. (2021)[P] • Toich, Schutt, and Fisher (2022)[P]
Careless responding "Respondents intentionally or unintentionally answer survey items in a manner that does not accurately reflect their true sentiments" (Ward and Pond 2015, p. 554).	• Oppenheimer et al. (2009)[P] • Meade and Craig (2012)[P] • Ward and Pond (2015)[P] • Weathers and Bardakci (2015)[M] • Niessen, Meijer, and Tendeiro (2016)[P]
Careless/insufficient effort (C/IE) responding "The individual responds without sufficient attention to the content and semantic polarity of the items" (Arias et al. 2020, p. 2490). "C/IE responders are missing data that is not actually missing. They have provided a response when they might have well left that response blank" (Curran 2016, p. 4).	• Weathers and Bardakci (2015)[M] • Curran (2016)[P] • Arias et al. (2020)[P]
Straightlining Respondents "provide similar responses regardless of the content and direction of the item" (Arias et al. 2020, p. 2490).	• Arias et al. (2020)[P]

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Table 1. Continued.

Term definition	Citations
Random responding "Respondent does not attend to the content of the item, but intentionally uses all response categories to appear to respond thoughtfully" (Arias et al. 2020, p. 2490).	• DeSimone and Harms (2018)[P] • Arias et al. (2020)[P]
Noncompliant responding "Individuals who show too much or too little variation in their responses" (Barends and de Vries 2019, p. 86).	• Barends and de Vries (2019)[P]
Stereotyped responding "A response that does not accurately represent subjects' attitudes" (Abbey and Meloy 2017, p. 63).	• Abbey and Meloy (2017)[G]
Response validity "The extent to which scores reflect the thoughts and beliefs of respondents concerning the phenomena of interest … If reported scores are aligned with the respondent's cognitions, then response validity is high, whereas if the scores are misaligned with what the respondent actually thinks, perceives, or feels, response validity would be low" (Edwards 2019, p. 63).	• Edwards (2019)[B]
Satisficing When some respondents take a shortcut and do not go through the mental process of coming up with an optimal answer (Krosnick 1991).	• Krosnick (1991)[P] • Conrad et al. (2017)[P] • Anduiza and Galais (2017)[P] • Liu and Wronski (2018)[P] • Morren and Paas (2020)[P]
Content nonresponsivity "[When] careless respondents give answers of bad data quality regardless of the content of the question" (Brühlmann et al. 2020, p. 2).	• Brühlmann et al. (2020)[P]

Note: Superscript capital letters refer to the primary academic discipline of the cited contribution:
[S] for Statistics.
[P] for Psychology.
[B] for Business.
[G] for Government/Political Science.
[M] for Marketing.

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nonresponsivity or "responding without regard to item content" (p. 438). Huang et al. (2012) popularized "insufficient effort responding"—a term now widely used in psychological research—which describes a type of response behavior "in which the respondent answers a survey measure with low or little motivation to comply with survey instructions, correctly interpret item content, and provide accurate responses" (p. 100). This type of response behavior includes both random and nonrandom careless responses. Other political scientists like Alvarez et al. (2019) also use an attention frame and raise concerns over inattentive respondents being a source of "non-sampling bias and response error" because these respondents are not "mindful in the survey process" (p. 146). Importantly, across definitions, there is a consensus over a conceptual distinction between inattentive responding and dishonest responding (or response distortion), with the latter treated as a different type of response behavior (Arthur, Hagen, and George 2021).

Whether researchers use "insufficient effort responding" or "inattention" to describe inattentive respondents, conceptually, they are referring to the same type of response behaviors. Though different fields have developed their own vocabularies to describe these behaviors, it is important to recognize the similarities in the underlying concepts being studied and not let semantic differences inhibit cross-field collaborations. In this review, consistent with Berinsky, Margolis, and Sances (2014), we will use the term "inattentive responding."

Measuring Inattention

A key consideration facing scholars is *how* to measure attention. This decision is crucial because incorrectly flagging attentive respondents as inattentive (Type I error) and failing to identify inattentive respondents as such (Type II error) can pose threats to the validity of the measures used by researchers (Curran 2016; Abbey and Meloy 2017).[2]

Direct Measures

Direct measures (sometimes referred to as attention checks or screeners) involve explicitly asking survey respondents a question or series of questions to determine some level of attentiveness. Common strategies include instructional manipulation checks (IMCs), bogus items, and instructed response items (IRIs). Newer methods include Factual Manipulation Checks (Kane

2. Moreover, the consequences of inattention likely differ across studies. In a survey experiment, for example, inattentive respondents in the treatment group may not have been actually "treated."

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Table 2. Direct measures of attentiveness.

Measure definition	Application	Citation
Instructional manipulation checks (IMC)		
"A question embedded within the experimental materials that is similar to the other questions in length and response format (e.g., Likert scale, check boxes, etc.). However, unlike the other questions, the IMC asks participants to ignore the standard response format and instead provide a confirmation that they have read the instruction" (Oppenheimer, Meyvis, and Davidenko 2009, p. 867).	"When a big news story breaks people often go online to get up-to-the-minute details on what is going on. We want to know which websites people trust to get this information. We also want to know if people are paying attention to the question. To show that you've read this much, please ignore the question and select ABC News and The Drudge Report as your two answers" (Berinsky, Margolis, and Sances 2014, p. 740).	• Berinsky, Margolis, and Sances (2014) • Hauser and Schwarz (2015) • Anduiza and Galais, (2017) • Paas, Dolnicar, and Karlsson (2018) • Mancosu, Ladini, and Vezzoni (2019) • Morren and Paas (2020)
Bogus item/trap question/infrequency scale		
"Items are constructed to appear face valid on a quick visual inspection, but obvious or absurd on deeper inspection" (Curran 2016, p. 13).	I was born on February 30th.	• Meade and Craig (2012) • Huang et al. (2012, 2015) • Liu and Wronski (2018) • Kim et al. (2018) • Curran and Hauser (2019)
Instructed response item		
"Items [that ask] respondents to provide a specific … [These] items evaluate respondents' compliance with simple and concise instructions" (Alvarez et al. 2019, p. 147).	"*For this item, select Response Option 5*. If respondents do not endorse the instructed response option, it is assumed they are being inattentive" (Beck, Albano, and Smith 2019, p. 375).	• Bowling et al. (2016) • Beck, Albano, and Smith (2019) • Alvarez et al. (2019) • Shamon and Berning (2020) • Berinsky et al. (2021) • Gummer, Roßmann, and Silber (2021)

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Table 2. Continued.

Measure definition	Application	Citation
Factual manipulation checks Asks respondents objective questions (i.e., questions with correct answers) about key elements of the experiment or about a study's contents (Kane and Barabas 2019, p. 234–35).	A study about student loan forgiveness asks, "According to the paragraph you just read, what is a potential consequence of the student loan forgiveness program?" (Kane and Barabas 2019, p. 240).	• Kane and Barabas (2019)
Mock vignette checks Feature content substantively similar to that of experimental vignettes and are followed by factual questions that gauge respondents' attentiveness to the mock vignette. It is a design-based technique that is meant to overcome the risk of posttreatment bias when trying to measure attention.	Brief vignette about an unrelated topic meant to be similar to the treatment, followed by a series of factual questions related to the content of the vignette.	• Kane, Velez, and Barabas (2023)
Response consistency Asking respondents the same question twice in different forms to see if answers vary.	Survey asked respondent's age twice in two separate formats, in the first question of the survey ("In what year were you born?") and in the last question ("What is your age?") (Liu and Wronski 2018, p. 35).	• Liu and Wronski (2018)
Response rounding Questions asking respondents to type in a numeric answer in a textbox.	On average, about how many hours do you watch television?	• Liu and Wronski (2018)

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Table 2. Continued.

Measure definition	Application	Citation
Knowledge question Questions based on common knowledge.	The logo for the Olympic Games is comprised of four interlocking rings (Response options: Yes, No) (Liu and Wronski 2018, p. 35).	• Liu and Wronski (2018)

and Barabas 2019) and Mock Vignette Checks (Kane, Velez, and Barabas 2023). See table 2 for a list of the various direct methods along with relevant articles that serve as good examples for each strategy.

While scholars have introduced a variety of direct measures, details about their implementation and best practices remain open questions. In an early paper on IMCs, Berinsky, Margolis, and Sances (2014) note that because we think of attentiveness as a latent construct, these types of questions are subject to measurement error like any other survey question. As a result, they recommend using multiple IMCs and aggregating responses into an attentiveness scale rather than relying on a single measure. In a later paper, Berinsky et al. (2021) investigate both standalone items along with items embedded within a grid (instructed response items and bogus items) and recommend the use of a multi-item scale that includes questions with both high and low passage rates. Other scholars concur, calling for the use of multiple screeners (Thomas and Clifford 2017; Kane, Velez, and Barabas 2023) with varying length and difficulty (Morren and Paas 2020).[3]

Of course, asking additional questions increases the survey length, thereby increasing the likelihood of inattentive responses and the cost of a survey. IMCs are standalone questions that take up more space in a questionnaire compared to bogus items (also known as infrequency items [Huang et al. 2015]) that can be inserted into a larger grid of questions. In fact, Liu and Wronski (2018) find that passing one attention check does not predict passing another one within the same survey and recommend only including a single question (though these authors stand apart from the field in advising against the use of multiple measures). Additionally, a series of articles have recommended using shorter and easier direct measures as opposed to longer ones that require a higher cognitive load (Anduiza and Galais 2017; Liu and

3. A challenge arises when using multiple items, as determining the appropriate threshold to identify a respondent as inattentive is not a straightforward process. See further discussion below.

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Wronski 2018; Mancosu, Ladini, and Vezzoni 2019; Ladini 2022). Mancosu, Ladini, and Vezzoni (2019) find that answer quality among respondents who pass an easy screener is similar to those who pass a medium or complex screener, recommending that researchers resort to simpler and shorter questions.

Another implementation concern involves the placement of an attention check. Liu and Wronski (2018) find no difference in passage rates between questions at the beginning or the end of a survey, while Reyes (2022) recommends placing items later in the survey. If the survey includes an experiment, it may be important to ensure that the direct measure comes before the treatment is administered to avoid the risk of posttreatment bias (Montgomery, Nyhan, and Torres 2018). Some argue that individual attention may wax and wane across a survey (Alvarez et al. 2019) so a direct measure should be asked as close as possible to the treatment, though Berinsky, Margolis, and Sances (2014) explicitly test this question and find no evidence of significant variation in respondent attention.

Taken together, the research has not yet reached a consensus on how best to implement direct attention measures—including the optimal number, difficulty, and placement within the survey. An important final note is that these considerations are not independent of one another. For example, if a researcher has the space to use multiple measures, then considerations about difficulty and placement will be different than if a researcher is only able to use a small number of measures.

Indirect Measures

Direct measures, however, are not the only way to measure attention on a survey. As opposed to direct measures, indirect measures do not involve explicitly asking respondents questions but instead are typically calculated post hoc, using information such as response patterns and response times. There are a wide variety of indirect response measures, and a few studies have done comprehensive jobs summarizing and comparing them (Maniaci and Rogge 2014; Curran 2016; Niessen, Meijer, and Tendeiro 2016; Abbey and Meloy 2017; DeSimone and Harms 2018; Leiner 2019; Goldammer et al. 2020; Brühlmann et al. 2020; Hong, Steedle, and Cheng 2020; Ward and Meade 2023). Table 3 provides a list of indirect measures along with brief descriptions and examples.

A key distinction between direct and indirect measures is that indirect measures do not require "correct" or "incorrect" answers to determine whether a respondent is flagged as inattentive. Instead, researchers rely on observed response behaviors (either response patterns or response times) believed to be indicative of inattention. Researchers then quantify the prevalence of these behaviors and assign a score to each respondent. They then

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Table 3. Indirect measures of attentiveness.

Measure definition	Applications
Response time approaches	
Post hoc method relying on survey completion times to detect inattentive responding.	• For a review, see (Matjašič et al. 2018). • Response time (Meade and Craig 2012; Weathers and Bardakci 2015) • Floodlight detection (Dogan 2018) • Response time attentiveness clustering (Read, Wolters, and Berinsky 2022) • Response time-based latent response mixture model (Ulitzsch et al. 2022)
Person-fit approaches	
Assessing the fit between an individual's response and the underlying measurement model to detect responses that are not consistent with the measurement model.	• Person-fit statistics (ex. Guttman errors) (Niessen, Meijer, and Tendeiro 2016) • Person-fit index (Beck, Albano, and Smith 2019) • Interactive cleansing method (Patton et al. 2019)
Response pattern approaches	
Identifying patterns of inconsistent or unreliable responding that suggest that a respondent is not paying attention or responding carefully.	• LongString index (Johnson 2005; Huang et al. 2012) • Response pattern indices (Huang et al. 2012) • Individual consistency/ inconsistency (Ward and Pond 2015) • Response variance (Weathers and Bardakci 2015) • Random response share (Malone and Lusk 2018) • Semantic synonyms/antonyms (DeSimone, Harms, and DeSimone 2015)
Outlier approaches	
Identifying and investigating individuals who are responding without sufficient effort that differs from their thoughtful counterparts in some way.	• Mahalanobis distance (Ward and Pond 2015)

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determine whether someone is attentive or not by comparing respondents' scores against set thresholds. A main benefit of this strategy is that indirect measures do not take up any real estate on a survey and are unobtrusive, meaning that survey respondents do not know their behaviors are being monitored. The main drawback of this strategy is that researchers have to make two critically important decisions. First, researchers must decide what type of response behaviors they deem deviant and ensure that the observed behavior effectively captures inattention. Second, researchers must select a cutoff threshold used to flag respondents as inattentive. We describe the current state of research regarding both decisions below.

Response Patterns

One common indirect approach involves analyzing individual response patterns from long question batteries. These pattern-based indirect measures are popular in psychological research, as respondents often complete long grids of questions. LongString analysis, in which researchers identify careless respondents by measuring the longest consecutive string of identical survey responses, is one response pattern strategy. This type of response behavior is also referred to as straightlining. The individual consistency approach is another strategy. Here, researchers identify respondents who provide contradictory or inconsistent responses within a long question battery. For a comprehensive review of different indirect measures and their application in psychology, see Ward and Meade (2023).

Studies that evaluate the effectiveness of multiple measures repeatedly find that while individual response pattern measures tend to be correlated, they do not all identify the same respondents as inattentive (Maniaci and Rogge 2014; DeSimone and Harms 2018). This should not be a surprise, as different methods are meant to identify different types of behavior. As mentioned earlier, LongString analysis can identify inattentive behavior that manifests through straightlining,[4] while individual consistency measures identify a different, less obvious response pattern. Researchers employing a response pattern strategy should therefore identify what sort of response pattern behavior is indicative of inattention in their specific study.

Response Times

Another type of indirect measure of attentiveness employs individual response times.[5] In brief, researchers identify outliers—those whose time spent

4. Inattentive behavior like straightlining can also lead to *inflated* scale reliability due to artificial correlation between scale items.

5. Matjašič et al. (2018) provide a review of the state of the literature on response times.

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on a survey is far different from the majority of respondents—to identify inattentive respondents on a survey. Importantly, inattentive respondents may either complete the survey too fast or too slow (Read, Wolters, and Berinsky 2022). Respondents may speed through a survey without fully comprehending questions, or respondents may take an extended time to complete the survey, all while not focusing on the content of the survey due to distractions. Furthermore, there is no consensus about the best way to measure response time (entire survey, page time, question time, etc.) (Greszki, Meyer, and Schoen 2015; Matjašič et al. 2018; Leiner 2019).

Additionally, some researchers question the exclusive use of response time measures. Matjašič, Vehovar, and Sendelbah (2021) find that many response time strategies are insufficient on their own when compared to direct measures and response-pattern-based indirect measures. Researchers interested in using response times must therefore decide whether they are concerned about rushing respondents, distracted respondents, or both; care about the time spent on the survey as a whole or specific parts of the survey; and whether measuring response time alone is a sufficient strategy for identifying inattentive respondents.

Flagging Respondents and Setting Appropriate Cutoffs

For both response time and response-pattern-based indirect measures, decisions over appropriate cutoff points are critically important because such decisions can significantly influence which respondents are classified as attentive (Meade and Craig 2012; Maniaci and Rogge 2014; Curran 2016; Wood et al. 2017; DeSimone and Harms 2018; Leiner 2019; Brühlmann et al. 2020; Goldammer et al. 2020; Chmielewski and Kucker 2020). If the cutoff is too restrictive, researchers inflate the number of inattentive respondents by falsely flagging attentive respondents as inattentive. If the cutoff is too relaxed, then only the most inattentive respondents may be flagged, artificially deflating the level of inattention in the sample.

With this concern in mind, scholars have proposed new strategies to determine these thresholds for both response times (Dogan 2018; Read, Wolters, and Berinsky 2022; Ulitzsch et al. 2022) and response patterns (Dunn et al. 2018; Kim et al. 2018; Patton et al. 2019; Yu and Cheng 2019; Schroeders, Schmidt, and Gnambs 2022). One example is Read, Wolters, and Berinsky (2022), who leverage per-question response times along with dimension reduction and an unsupervised clustering algorithm to classify attentive respondents without having to specify a specific speed threshold. Other examples include Ulitzsch et al. (2022) and Arias et al. (2020), who both propose model-based methods leveraging mixture models that do not depend on researcher-defined cutoffs. Another approach proposed by Harden, Sokhey, and Runge (2019) uses response times as a measure of treatment

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compliance in surveys. This approach sidesteps the threshold discussion by conceptualizing inattention as a compliance problem. In doing so, researchers can estimate the causal average compiler effect (CACE) among attentive respondents.[6]

Decisions over cutoffs and flagging respondents are further complicated when scholars choose to use multiple indirect measures, as implementation is not straightforward (Meade and Craig 2012; DeSimone and Harms 2018). Due to the wide selection of indirect measures available, it is not always clear what the optimal combination of measures is. Moreover, different measures each have their own cutoff scores and may identify different sets of respondents. One recommendation for using multiple measures is the "multiple hurdle" approach, where different methods are deployed sequentially to identify inattentive respondents (Curran 2016; Goldammer et al. 2020). Brühlmann et al. (2020) warn this approach may be too restrictive and lead to false positives. Additionally, there is concern over the aggregation of multiple indices that involve user-defined cutoffs as opposed to more objective indicators used in direct measures (Maniaci and Rogge 2014; Brühlmann et al. 2020). A different approach involves calculating multiple indirect measures, but then identifying careless respondents using the "best subset" of measures that optimizes a balance of sensitivity and specificity across different careless responding behaviors (Hong, Steedle, and Cheng 2020).

All told, the research suggests that scholars should think carefully about what sorts of inattentive behaviors pose the greatest threats to their study and what kind of direct or indirect measure is most appropriate to identify respondents engaging in this behavior. Moreover, researchers should consider the empirical consequences of employing different cutoffs, different modeling approaches, and multiple measures of attentiveness.

Can We Increase Attentiveness?

In addition to trying to measure attention, researchers can attempt to increase or encourage attention through the use of warnings or instructions. These strategies forgo identifying a specific type of response behavior, and instead rely on inducing more attentive responses through the survey design. As an example, researchers may include a warning to respondents that their answers will be reviewed for quality control. The evidence in support of using warnings through survey instructions to increase attention is mixed (Meade and Craig 2012; Clifford and Jerit 2015; Ward and Pond 2015; Breitsohl and Steidelmüller 2018; Paas, Dolnicar, and Karlsson 2018; Ward and Meade 2018; Shamon and Berning 2020; Bowling et al. 2021; Toich, Schutt, and

6. This approach does, however, add additional modeling assumptions associated with two-stage least squares estimation.

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Fisher 2022). A related approach involves introducing virtual proctors that encourage attentive responding, though evidence in support of this approach is limited (Francavilla, Meade, and Young 2019). Researchers can also incorporate warnings with certain indirect measures like response time. For example, Conrad et al. (2017) implement an interactive prompting technique where respondents who answered faster than a certain response time threshold received a message encouraging them to answer carefully and to take their time. An extension of this strategy involves explicitly "training" respondents who fail attention checks through the use of additional instructions. While training respondents may increase attention check passage rates, Berinsky, Margolis, and Sances (2016) find that training does not increase general attentiveness throughout the survey. Importantly, these strategies used to increase attention are not mutually exclusive from the other measurement approaches discussed earlier. Future work should explore how a combination of both approaches can be used to improve data quality, encouraging attentiveness upfront while retaining traditional measures for identifying inattentive respondents when necessary.

Validation Strategies

When measuring attention, researchers should ensure that the strategy chosen appropriately captures the construct of interest. As a result, it is imperative for researchers to employ high-quality measures. For most researchers, this simply means selecting measurement strategies that have already been carefully validated. Scholars interested in developing new measurement strategies, on the other hand, ought to devote significant effort to show that their proposed measures are valid and reliable. While the number of novel methods to measure survey attention has grown, there is a lack of common validation strategies that facilitate comparison across different methods.

When developing new methods, the choice of appropriate validation approach depends on whether the proposed measurement strategy is direct or indirect.[7] A recent study by Kane, Velez, and Barabas (2023) introduces a novel direct measure called "Mock Vignette Checks," providing a compelling example of proper validation. This new measurement strategy relies on using a pretreatment "mock" experimental vignette that approximates the true treatment as a measure of attention. The authors employ a variety of validation strategies to test the construct and convergent validity of the measure. First, the authors replicate multiple experiments with known treatment effect sizes to evaluate treatment effect heterogeneity across levels of attention. This strategy is common, as it allows researchers to compare results to past studies that have

7. This difference is due to the distinctions in how those questions measure inattention, as discussed earlier.

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already been replicated (Berinsky, Margolis, and Sances 2014; Greszki, Meyer and Schoen 2015; Brühlmann et al. 2020). In addition to replication, the authors also compare mock vignette checks to other validated direct measurement strategies such as instructional manipulation checks along with a latent measure of attention such as response time. Importantly, these tests are conducted across multiple different sample providers.

Another way researchers have validated their direct measures of attention is by examining responses to factual survey questions that can be compared to external administrative data (Clifford and Jerit 2015; Alvarez and Li 2023). Alvarez and Li (2023) used a survey with validated voter turnout to assess the accuracy of the direct attentiveness measures. Using the behavioral data, the authors find that inattentive respondents are less likely to provide accurate reports of their voting history.

Some measurement strategies, such as response-time-based approaches, are especially difficult to validate due to the possibility that inattentive respondents may complete a survey either too *quickly* or too *slowly*. The literature on response-time-based approaches generally lacks consensus on appropriate validation strategies (Meade and Craig 2012; Conrad et al. 2017; Dogan 2018; Matjašič et al. 2018; Ulitzsch et al. 2022). In a recent paper, Read, Wolters, and Berinsky (2022) proposed a new measure called response-time attentiveness clustering (RTAC), which leverages dimension reduction and an unsupervised clustering algorithm to identify inattentive respondents. To validate this new measure, the authors used three strategies which included comparing open-ended questions across different response times, replicating a well-known survey experiment, and using a flipped scale in a series of ideological questions. To evaluate differences in scale reliability across levels of attention, the authors used Cronbach's alpha, a common tool used to validate indirect measures that rely on long scales (Hauser and Schwarz 2015; Ladini 2022).

There are a few important takeaways here. For practitioners interested in measuring attention, it is essential to only employ properly validated measures. On the other hand, methodologists interested in proposing new measures must spend significant effort validating these measures. When doing so, researchers must ensure that the strategies employed align with the goal of the measurement strategies being tested. Additionally, the validation strategies should ensure both construct and convergent validity. Finally, an undervalued component of developing a new measure involves making it easy for practitioners to correctly implement the validated method. Providing practitioners with ready-made example questions can facilitate the appropriate implementation of the measurement strategy (Berinsky, Margolis, and Sances 2014; Kane, Velez, and Barabas 2023).

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Who Is Inattentive and Why?

Aggregate rates of survey inattention vary widely across studies and depend on a multitude of factors related to researcher decisions, survey context, and respondent behavior. First, researcher decisions can influence passage rates at multiple points in the survey process. Most notably, the choice of measurement strategy has a notable impact on flagging inattentive respondents, as different strategies may identify different respondents (Maniaci and Rogge 2014; DeSimone and Harms 2018). As discussed earlier, researchers can also affect passage rates when using both direct and indirect measures through choices related to cutoff thresholds and decisions over question construction such as the length, difficulty, and placement of an item. If a researcher chooses a difficult direct measure, a greater number of respondents will be flagged as inattentive.

Second, the choice of sample provider used can affect levels of attention (Abbey and Meloy 2017; Brühlmann et al. 2020). Researchers have found variation in attentiveness rates across sample providers such as MTurk (Chmielewski and Kucker 2020) and Lucid (Ternovski et al. 2022), as well as comparisons between samples (Hauser and Schwarz 2016; Arndt et al. 2022). Differences in sample screening methods across providers produce variations in sample quality, which can impact sample attentiveness. Sample quality may also fluctuate over time for a particular sample provider, resulting in temporal variation in passage rates (Ternovski et al. 2022; Peyton, Huber, and Coppock 2022).

Third, scholars have explored how respondent-level characteristics correspond with individual-level attention. Evidence on the relationship between personality traits such as agreeableness or consciousness and attentiveness is mixed (Maniaci and Rogge 2014; Bowling et al. 2016; Palaniappan and Kum 2019). Bowling et al. (2016) argue that insufficient effort responding is a reflection of personality traits and find evidence in support of this claim—over time, individual levels of inattention were consistent, reflecting what they call "enduring individual differences" (Bowling et al. 2016). Other scholars disagree and find suggestive evidence that inattention may operate independently of personality traits and call for additional research on the subject that incorporates both individual traits along with survey context-specific factors, such as the relevance of the survey to respondents, attention capacity, and other contextual factors (Maniaci and Rogge 2014; Palaniappan and Kum 2019). This research can build on past work linking survey attention to motivation (Rios et al. 2017; Bowling et al. 2021). Of interest is a potential interaction between personality and context where individuals with certain personality traits might be less likely to respond carefully if the survey is of little interest to them. Additionally, this question of internal motivation is also tied to the survey-taking *context*. For example,

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an individual's capacity to engage with a survey will likely vary depending on whether they take it uninterrupted at home or during their commute on public transportation. While no existing work directly examines this question, insights about the role of survey *design* in influencing attention can likely extend to the role of the survey-taking context.

Additionally, some studies have found demographic differences between attentive and inattentive respondents in terms of education, gender, and age (Berinsky, Margolis, and Sances 2014; Anduiza and Galais 2017; Paas, Dolnicar, and Karlsson 2018; Alvarez et al. 2019; Mancosu, Ladini, and Vezzoni 2019). Other studies have found consistent differences along age and race lines. In general, studies have found that whiter, older, and more educated respondents are more likely to be identified as attentive. While these differences tend to be substantively small, they raise concerns about how adjustments for inattention may affect external validity (Thomas and Clifford 2017; Alvarez et al. 2019; Kane, Velez, and Barabas 2023).

These findings further reinforce the importance of transparency throughout the research process. Since attentiveness measures are predicated, in part, on how researchers measure and determine attention, researchers must be clear about the choices they make in this arena. Moreover, potential differences between samples and demographic disparities between attentive and inattentive respondents underscore the importance of providing comprehensive details about the data collection process and the demographic characteristics of the attentive group.

Handling Inattentive Respondents

Following decisions about how to measure attentiveness on a survey, researchers must then consider the appropriate approach to handle the respondents identified as inattentive. If inattentive respondents are viewed as random noise, the easy way to handle them would be to drop them from the sample. Alvarez et al. (2019) present a useful framework for handling inattentive respondents and discuss four different strategies. These strategies include doing nothing and keeping all respondents, dropping inattentive respondents, dropping and reweighting, and keeping all respondents and accounting for inattention through a model-based adjustment. Alvarez et al. (2019) review the limitations of each of these approaches and the situations in which they would be appropriate, highlighting the need for researchers to carefully consider the trade-offs associated with these decisions and consider how choices made align with the research goals. Alvarez and Li (2023) offer further guidance for researchers, suggesting they examine whether survey results are sensitive to attention levels and if attentiveness correlates with the outcome of interest. If either of these conditions holds, then dropping respondents is likely not appropriate.

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Removing inattentive respondents can improve scale reliability, increase precision, reduce bias, and increase statistical power by reducing noise (Hauser and Schwarz 2015; Abbey and Meloy 2017; Thomas and Clifford 2017; Hong, Steedle, and Cheng 2020; Gummer, Roßmann, and Silber 2021; Pyo and Maxfield 2021). But under what conditions can these respondents be considered true noise? Importantly, no single rule dictates when dropping respondents is acceptable and caveats exist regarding the type of sample used and the different conditions that make removal acceptable.

Many scholars argue that researchers should not drop inattentive respondents because it could bias the estimates (Berinsky, Margolis, and Sances 2014; Ward and Pond 2015; Anduiza and Galais 2017; Mancosu, Ladini, and Vezzoni 2019; Tyler, Grimmer, and Westwood 2022; Atsusaka and Stevenson 2023). If inattention is correlated with theoretically relevant demographic characteristics, dropping respondents can introduce bias in the sample. Additionally, removing respondents may significantly reduce the size of the sample, reducing statistical power. Silber, Danner, and Rammstedt (2019) argue that removing inattentive respondents only marginally increases data quality, so it may not be worth the potential risks. DeSimone and Harms (2018) make a similar argument. One common recommendation is to create an additive index of attention and report results at multiple levels of attention (Berinsky, Margolis, and Sances 2014; Alvarez and Li 2023). Ultimately, researchers need to be explicit about their choices and, if they choose to drop inattentive respondents, they should report results among both the full sample and the attentive sample, along with their demographic compositions. These practices maximize transparency and allow readers to evaluate how robust the results are to inattention along with adjustments for inattention.

In a similar vein, to justify the exclusion of inattentive respondents, Buchanan and Scofield (2018) conducted a sensitivity analysis where they varied the proportion of the sample that was flagged as inattentive to better understand the consequences of removing these respondents. They find that keeping respondents flagged as inattentive can increase noise in the data and reduce the study's statistical power. In a working paper, Tyler, Grimmer, and Westwood (2022) propose a new statistical framework that allows for the identification of population attitudes and treatment effects from surveys that contain a mix of attentive and inattentive respondents without the risk of having to drop respondents. Leveraging cutting-edge machine-learning and weighting techniques, the authors provide an example of statistical methods for handling inattention, a promising area for future research.

Finally, researchers should assess the possible implications of striving for a fully attentive sample. For example, consider a study on the persuasiveness of different frames of a campaign ad. In the real world, many individuals

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exposed to campaign messaging may not actually pay attention to the ads. To more accurately reflect the continuum of attention in the broader public, it may be necessary to retain inattentive respondents in the sample, as removing inattentive respondents may result in inflated treatment effect estimates. On the other hand, if the experimental treatment is simply a means to induce variation in something else, say emotion, inattentive respondents may contribute nothing to our ability to learn about the effects of emotions and a researcher may be justified in adjusting for inattention in some way. This example highlights the importance of researchers considering the desired attention level within a sample.

Potential Risks of Measuring Attention

When determining what strategies to use to identify inattentive respondents, researchers should also consider the possibility that in addition to measuring attention, these strategies can potentially influence respondent behavior in unintended ways. Such risks are especially relevant when it comes to direct measures. Concerns about the unintended effects of attention checks connect to broader questions of survey design. As discussed earlier, researchers need to be careful when designing direct attention checks and consider trade-offs related to the difficulty, length, and placement of the question. Ideally, researchers should aim to implement attention checks that capture inattentive respondents without altering respondent behavior. Hauser and Schwarz (2015) argue that IMCs can impact systemic thinking, though Hauser and Schwarz (2016) later add an important caveat: this concern may only apply to certain complex reasoning tasks, not survey taking in general. Consequently, researchers must consider whether this threat is relevant to their specific research questions and survey instrument. In a review, Abbey and Meloy (2017) also warn of possible psychological consequences of obstructive attention checks that may generate unintended emotional responses. It should be noted that the consequences of attention checks are not necessarily purely negative; Shamon and Berning (2020) find that certain direct measures do not have demotivating effects on response behavior, but instead have a motivational influence on respondents by increasing cognitive effort. While inducing attentive responses through the use of an attention check may improve data quality, researchers must also be mindful of the possibility of artificially inflating attentiveness within the sample.

A growing body of research has found that most direct measures seem to have little effect on respondent behavior—either positive or negative. There is little evidence that direct measures bias survey responses or threaten scale validity (Kung, Kwok, and Brown 2018; Paas, Dolnicar, and Karlsson 2018; Gummer, Roßmann, and Silber 2021), lead to survey drop-off (Paas and Morren 2018), or deteriorate the survey-taking experience (Breitsohl and Steidelmüller 2018).

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These findings apply across a variety of direct methods (Huang et al. 2015; Paas, Dolnicar, and Karlsson 2018; Gummer, Roßmann, and Silber 2021). Encouragingly, Kane, Velez, and Barabas's (2023) recent work finds that the effect of including a mock-vignette check on non-attention-related response behavior is close to zero.

While the growing evidence affirming the lack of negative consequences regarding attention checks further solidifies the value of these strategies, future work should explore the effects of variation in both the difficulty and number of attention checks on survey behavior. Gummer, Roßmann, and Silber (2021) express concern over possible spillover effects due to too many attention checks. Additionally, researchers should continue to explore the relationship between attention checks and different types of survey tasks, ranging from simple survey questions to more complex reasoning tasks (Hauser and Schwarz 2015; Kung, Kwok, and Brown 2018). Researchers may also wish to avoid recycling common questions, as certain sample providers may have subsets of experienced survey takers who may be familiar with attention checks, pass them, and then continue to provide inattentive, low-quality answers to the rest of the survey (Barends and de Vries 2019).

When thinking about the unintended consequences of measuring attention, researchers should keep a few other points in mind. Curran and Hauser (2019) urge researchers to account for the threat of false positives when using bogus or infrequency items. Liu and Wronski (2018) also find evidence of both false positives and false negatives when using instructed response items. When conducting cross-national research, Grau, Ebbeler, and Banse (2019) argue for caution when measuring attention due to the possibility of cultural sources of variation in attention. Alvarez and Li (2023) also discuss the decision between different types of direct measures as a sort of bias/variance trade-off. They see IMCs as a more aggressive measure that may reduce bias in survey estimates, but introduce variance by flagging respondents who may still provide some attentive responses.

Conclusion

In this review, we examine a wide variety of strategies for measuring survey attention and debates over best practices when implementing these strategies. Scholars working within this literature have often focused on narrow research questions at the expense of developing a higher-level set of recommendations aimed toward helping public opinion researchers understand the central considerations when measuring attention and addressing inattentive respondents. This is evident in the use of varying terminology across disciplines to describe essentially identical classes of response behaviors. Rather than siloing small advances within different academic disciplines, future

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research on survey attention should strive for interdisciplinary collaboration, focusing on a common goal of improving survey data quality.

Throughout this piece, we stress the importance of survey researchers weighing multiple considerations when deciding how to measure inattention. Additionally, we emphasize the need for greater transparency when measuring survey attention. Researchers need to adopt a holistic approach to measuring attention, thinking deeply about how their research objectives relate to the significance of attentive respondents and how different measurement strategies might affect those goals. Factors such as the sample provider, survey content, and survey design should all influence researchers' decisions on how to address inattention.

Researchers must also exercise caution when flagging respondents as inattentive, as decisions over the difficulty of an attention check or appropriate cutoff threshold will affect passage rates. It is crucial to remember that inattention is a property of both the individual and the survey context. Insufficient effort responding or inattention describes a specific type of response behavior, not a specific type of respondent, and can be induced by both individual-level characteristics as well as survey context.

When reporting the results of the survey, researchers need to disclose information about how they chose to measure attention. This includes information about what measures were used, and who was flagged as inattentive, along with researcher choices regarding cutoffs, if applicable. Additionally, if analyzing a subset of attentive respondents, researchers should report results among both the full sample and the attentive sample along with the demographic characteristics of each sample. To facilitate this process, we encourage researchers to include information about how survey attention will be measured and how inattentive respondents will be handled when preregistering a survey or a survey experiment. Such practices will increase transparency when it comes to measuring attention and help researchers better understand the implications of different measurement strategies while minimizing the threat of *p*-hacking due to false positives among attentive and inattentive respondents.

Although the literature on survey inattention may be overwhelming, we have witnessed significant progress in recent years and there are some promising directions for future work. First, the field should focus on developing comprehensive recommendations for implementing both direct and indirect measures of attentiveness. This includes questions about the placement and difficulty of direct measures as well as suitable cutoff thresholds for indirect measures. A recent paper by Berinsky et al. (2021) is a good model for this type of work and provides concrete recommendations about the number and the level of difficulty of attention checks that should be used. At the same time, many ongoing debates remain unresolved, leaving room for innovation—especially regarding the question of cutoffs on indirect measures.

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Second, additional work should explore the relationship between individual-level traits, survey context, and survey inattention using longitudinal data (Paas, Dolnicar, and Karlsson 2018; Goldammer et al. 2020; Gummer, Roßmann, and Silber 2021). Variations in survey context may interact with individual-level predispositions to induce different response behaviors in distinct survey waves. Third, while evidence suggests that the behavioral consequences of direct measures are limited, the effects of these measures may vary depending on the type of survey question (Hauser and Schwarz 2015; Kung, Kwok, and Brown 2018). Moreover, as we call for the use of multiple attention checks within a survey when possible, researchers should also investigate the potential behavioral consequences of both increasing the number and the level of difficulty of these measures. Fourth, scholars should continue leveraging state-of-the-art statistical modeling techniques to explore solutions for setting appropriate cutoff thresholds for indirect measures and removing inattentive respondents. Finally, concerns over survey inattention gained prominence as surveys shifted from primarily phone-based to online formats. In recent years, survey respondents have increasingly completed surveys online using their mobile phones instead of their computers. Phones and computers are fundamentally different devices, necessitating work on how survey inattention manifests on this newer medium.

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